$m_1 \times m_2 = -1$

GRADIENT EXERCISE:

1. Determine the gradient of the following lines with A and B points on the line:

a) A(3; 2) and B(5; 4) 1

b) A(5; 8) and B(7; 16) 4

c) A(4; 9) and B(6; -1) $\frac{-10}{2} = -5$

d) A(-3; 5) and B(-8; -5) 2

e) A(4; 7) and B(-3; -2) $\frac{-9}{-7} = \frac{9}{7}$

f) A(-8; -3) and B(-2; -9) -1

2. Write down the gradient of the line perpendicular to each line in question 1.

a) -1 b) $-1/4$ c) $1/5$ d) $-1/2$ e) $-7/9$ f) 1

3. Are the following lines (AB and CD) parallel, perpendicular or neither?

a) A(-3; 5); B(5; -1) $-3/4$ and C(-2; -1); D(1; 3) $4/3$ → ⊥

b) A(-2; 4); B(3; 1) $\frac{3}{-5}$ and C(5; -1) and D(-2; 6) $\frac{-7}{7} = -1$ → neither

c) A(-2; 1); B(2; 4) $3/4$ and C(-3; -1) ; D(0; 3) $\frac{4}{3}$ → neither

4. Calculate the value of x if A(-2; 3) B(1; 4) C(-4; 1) D(x; 4) and

a) AB // CD $x = 5$

b) AB ⊥ CD $x = -5$

See below

(4) A(-2 : 3) B(1 : 4) C(-4 : 1) D(x 4)

a) AB // CD $m_1 = m_2$

$m_{AB} = \frac{4-3}{1-(-2)} = \frac{1}{3}$

$\therefore m_{CD} = \frac{1}{3} = \frac{4-1}{x-(-4)}$

$\frac{1}{3} = \frac{3}{x+4}$ cross multiply

$(x+4) = 9$

$x = 5.$

b) AB ⊥ CD $m_1 \times m_2 = -1$

$m_{AB} = \frac{1}{3}$ $m_{CD} = -3$

$m_{CD} = \frac{-3}{1} = \frac{3}{x+4}$

$-3(x+4) = 3$

$-3x - 12 = 3$

$-3x = 15$

$x = -5$